Data Appendix

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DS 4002: Project 2

Section 1: NY Air Quality (Uncleaned Dataset)

1.1 Unit of Observation

Each row represents one measurement of New York air quality taken between Jan. 1 2005 - Jun. 1 2023.

1.2 Dataset Description

This dataset contains the original Air Quality metadata and titles. It includes 18,862 rows and 12 columns: Unique ID, Indicator ID, Measure, Measure Info, Geo Type Name, Geo Join ID, Geo Place, Time Period, Start_Date, Data Value, and Message.

1.3 Data Variable Explanations

Column Name	Description	Data Type	Example Value
Unique ID	Unique identifier for each measurement	Integer	336867
Indicator ID	Numeric code identifying the specific air quality indicator	Integer	375
Name	Name of pollutant/ air quality indicator measured	String	“Nitrogen dioxide (NO2)”
Measure	Measurement taken and being reported	String	“Mean”
Measure Info	Unit of measurement	String	“ppb”
Geo Type Name	Type of geographic unit (boundary)	String	“CD”
Geo Join ID	Numeric geographic identifier to join with spatial datasets	Integer	407

Geo Place Name	Name of geographic unit (place)	String	“Flushing and Whitestone (CD7)”
Time Period	General time period measurement was taken (multi-month range)	String	“Winter 2014-15”
Start_Date	Start date of measurement period	Date/String	“1/1/2017”
Data Value	Actual air quality measure for that location & indicator	Float	23.97

1.4 Statistics

- Total number of air quality measurements: 18,862
- Data range covered: 18.5 years
- Number of community districts: 18
- Number of air quality indicators: 5
- Number of different geographic levels: 4

1.5 Figures

- N/A

Section 2: NY Air Quality (Cleaned Dataset)

2.1 Unit of Observation

Each row represents one measurement of New York air quality taken between Jan. 1 2005 - Jun. 1 2023.

2.2 Dataset Description

This dataset contains the NY air quality measurements post cleaning. It includes 18,862 rows and 13 columns: Unique ID, Indicator ID, Measure, Measure Info, Geo Type Name, Geo Join ID, Geo Place, Time Period, Start_Date, Data Value, and Message.

2.3 Data Variable Explanations

Column Name	Description	Data Type	Example Value
Unique ID	Unique identifier for each measurement	Integer	131263
Indicator ID	Numeric code identifying the specific air quality indicator	Integer	657
Name	Name of pollutant/ air quality indicator measured	String	“Asthma emergency department visits due to PM2.5”
Measure	Measurement taken and being reported	String	“Estimated annual rate (age 18+)”
Measure Info	Unit of measurement	String	“Per 100,000 adults”
Geo Type Name	Type of geographic unit (boundary)	String	“Borough”
Geo Join ID	Numeric geographic identifier to join with spatial datasets	Integer	2

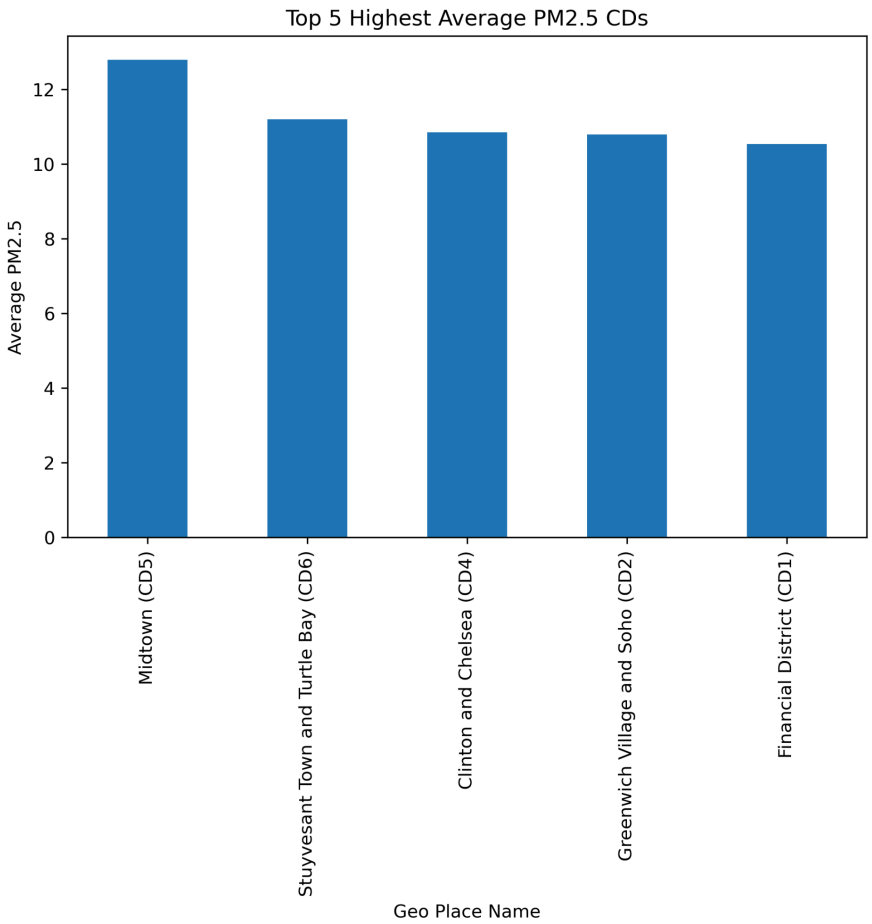
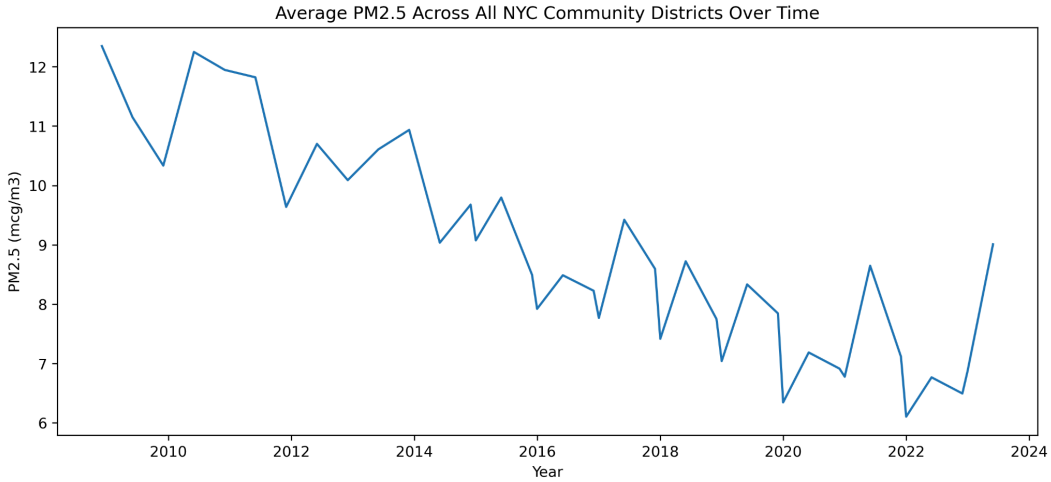
Geo Place Name	Name of geographic unit (place)	String	“Brooklyn”
Time Period	General time period measurement was taken (multi-month range)	String	“2005-2007”
Start_Date	Start date of measurement period	Date/String	“1/1/2005”
Data Value	Actual air quality measure for that location & indicator	Float	55.5
Message	Additional notes about measurement	String	#######

2.4 Statistics

- Total number of air quality measurements: 18,862
- Data range covered: 18.5 years
- Number of community districts: 18
- Number of air quality indicators: 5
- Number of different geographic levels: 4

2.5 Figures

- Top 5 Highest Average PM2.5 CDs



- Average PM2.5 Across All NYC Community Districts Over Time